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Ammunition magazine pouch and methods of assembling same

Abstract

An ammunition magazine pouch is described herein. The ammunition magazine pouch includes a magazine support assembly including an interior surface defining a magazine chamber extending between an open top end and a closed bottom end. The magazine chamber is sized and shaped to receive at least one ammunition magazine therein. The magazine support assembly includes a front cover member including an inner surface defining a portion of the magazine chamber, a back plate member coupled to the front cover member to define the magazine chamber therebetween, and a tensioning assembly coupled to the front cover member and the back plate assembly for adjusting a distance between an upper portion of the front cover member and an upper portion of the back plate member to adjust a width of the open top end of the magazine chamber.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/275,266, filed on Nov. 3, 2021, the disclosure of which are hereby incorporated by reference in its entirety for all purposes.

FIELD OF THE INVENTION

(1) This invention relates generally to an ammunition magazine pouch and a molding assembly for forming the ammunition magazine pouch.

BACKGROUND OF THE INVENTION

(2) Law enforcement personal typically wear duty belts with equipment that includes but is not limited to a holstered handgun, holstered taser, holstered pepper-spray, and spare ammunition. Law enforcement personal are required to carry spare ammunition magazines on their duty belt. The device for holding the spare ammunition magazines in this case are critical because they can limit the speed of access to the magazines when re-loading the handgun. However, the size of the case must be as small as possible so that the positioning on the duty belt is comfortable for the police officer.

(3) The present invention addresses one or more of the aforementioned challenges.

SUMMARY OF THE INVENTION

(4) In one aspect of the present invention, an ammunition magazine pouch is provided. The ammunition magazine pouch includes a magazine support assembly including an interior surface defining a magazine chamber extending between an open top end and a closed bottom end. The magazine chamber is sized and shaped to receive at least one ammunition magazine therein. The magazine support assembly includes a front cover member including an inner surface defining a portion of the magazine chamber, a back plate member coupled to the front cover member to define the magazine chamber therebetween, and a tensioning assembly coupled to the front cover member and the back plate member for adjusting a distance between an upper portion of the front cover member and an upper portion of the back plate member to adjust a width of the open top end of the magazine chamber.

(5) In another aspect of the present invention, a method of manufacturing an ammunition magazine pouch is provided. The pouch includes an interior surface defining a magazine chamber configured to receive an ammunition magazine therein. The method includes thermoforming a front cover member from a first sheet of thermoplastic material and thermoforming a back plate member from a second sheet of thermoplastic material. The method also includes stitching the back plate member to an attachment assembly and coupling the front cover member to the back plate member to define the magazine chamber therebetween extending between an open top end and a closed bottom end. The method also includes coupling a tensioning assembly to the front cover member and the back plate member to adjust a distance between an upper portion of the front cover member and an upper portion of the back plate member to adjust a width of the open top end of the magazine chamber.

(6) In a further aspect of the present invention, a system for manufacturing an ammunition magazine pouch is provided. The system includes a thermoplastic machine assembly including a support frame, a heat press machine mounted to the support frame, a vacuum forming machine assembly mounted to the support frame adjacent to the heat press machine, and a support sled slideably coupled to the support frame and movable between the heat press machine and the vacuum forming machine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Other advantages of the present invention will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:
- (2) FIGS. **1A-20** are perspective views of an ammunition magazine pouch, according to embodiments of the present invention;
- (3) FIGS. **21** and **22** are exploded perspective views of the ammunition magazine pouch including a magazine support assembly and an attachment system;
- (4) FIGS. **23-33** are perspective views of portions of the magazine support assembly including a front cover member and a back plate member;
- (5) FIGS. **34-42** are perspective views of the back plate member and the attachment system;
- (6) FIGS. **43-46** are perspective views of a thermoplastic machine assembly for use in manufacturing the ammunition magazine pouch, according to embodiments of the present invention;
- (7) FIGS. **47-48** are perspective views of a CNC milling machine for use in manufacturing the ammunition magazine pouch, according to embodiments of the present invention;
- (8) FIGS. **49-62** are perspective views of mold assemblies for use in manufacturing the ammunition magazine pouch, according to embodiments of the present invention; and
- (9) FIG. **63** is a flowchart illustrating a method of manufacturing the ammunition magazine pouch, according to embodiments of the present invention.
- (10) Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION OF THE INVENTION

- (11) In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent, however, to one having ordinary skill in the art that the specific detail need not be employed to practice the present invention. In other instances, well-known materials or methods have not been described in detail in order to avoid obscuring the present invention.
- (12) In general, the present invention describes a tactical pouch **10** for use in securing tactical gear and ammunition magazines to traditional, micro and laser cut Molle attachment systems as well as slide on tactical and/or utility belts. For example, the tactical pouch **10** may be used to hold tactical gear such as CAT Tourniquets, telescoping batons, and/or any suitable tactical gear. The tactical pouch **10** may also be used to hold ammunition magazines such as, for example, .40, 9 mm, and .357 sig double stack magazines, AR15 magazines, and/or any suitable ammunition magazine.
- (13) Referring to FIGS. **1A-42**, in the illustrated embodiment, the present invention includes an ammunition magazine pouch **10** that includes a magazine support assembly **12** and an attachment system **14**. The magazine support assembly **12** includes an interior surface that defines a magazine chamber **16** extending between an open top end **18** and a closed bottom end **20**. The magazine chamber **16** is sized and shaped to receive at least one ammunition magazine **22** therein. The magazine support assembly **12** includes a front cover member **24** including an inner surface defining a portion of the magazine chamber **16**, a back plate member **26** coupled to the front cover member **24** to define the magazine chamber **16** therebetween, and a tensioning assembly **28** coupled to the front cover member **24** and the back plate member **26**. The tensioning assembly **28** is configured to adjust a distance between an upper portion of the front cover member **24** and an upper portion of the back plate member **26** to adjust a width of the open top end of the magazine chamber **16**. In some embodiments, the ammunition magazine pouch **10** is sized to hold the plurality of ammunition magazines side by side, for example, the left side of one of the ammunition magazines faces the right side of the second magazine.

(14) Referring to FIGS. 23-29, in the illustrated embodiment, the front cover member 24 includes a pair of opposing sidewalls 30 extending outwardly from a front endwall 32, a bottom endwall 34 extending between the front endwall 32 and the opposing sidewalls 30, a support flange 36 extending outwardly from the opposing sidewalls 30 and the bottom endwall 34, and a pair of opposing raised fastening support members 38 extending outwardly from the opposing sidewalls 30 defined near the open top end of the magazine support assembly 12. The front cover member 24 is coupled to the back plate member 26 such that the support flange 36 contacts a front surface of the back plate member 26. In some embodiments, the support flange 36 of the front cover member 24 is coupled to the front surface of the back plate member 26 with an adhesive. In some embodiments, the front cover member 24 is formed of a unitary body including a thermoplastic material.

(15) Each raised fastening support member 38 includes an inner surface spaced a distance from the back plate member 26 to define a gap 40 between the inner surface of the raised fastening support member 38 and the back plate member 26.

(16) In some embodiments, the front cover member 24 includes a pair of recessed channels 42 defined along an outer surface of the front cover member 24 and extending from the front endwall 32 to the bottom endwall 34. Each recessed channel 42 defines a corresponding dividing ridge 44 extending into the magazine chamber 16 to define a plurality of magazine compartments 46 within the magazine chamber 16. Each magazine compartment 46 is sized and shaped to receive a corresponding ammunition magazine 22 therein. In some embodiments, the front cover member 24 includes a plurality of drain openings 48 spaced along the bottom endwall 34.

(17) Referring to FIGS. 30-34, in the illustrated embodiment, the back plate member 26 includes a support plate 50 having a substantially rectangular cross-sectional shape extending between a front surface 52 and rear surface 54. The front surface 52 defines a portion of the magazine chamber 16 and includes a pair of raised positioning members 56 extending into the magazine chamber 16. Each raised positioning member 56 is orientated with respect to a corresponding dividing ridge 44 of the front cover member 24 and spaced a distance from the corresponding dividing ridge 44 to define the plurality of magazine compartments 46. The support plate 50 may also include an arcuate surface 58 extending along a top portion of the support plate 50 to define the open top end of the magazine chamber 16. In some embodiments, the back plate member 26 is formed of a unitary body including a thermoplastic material.

(18) Referring to FIGS. 1A-20, in the illustrate embodiment, the tensioning assembly 28 includes a plurality of adjustable fastening members 60 coupled between the front cover member 24 and the back plate member 26. The plurality of adjustable fastening members 60 includes a pair of opposing outer fastening members 62 coupled to the opposing raised fastening support members 38 and the back plate member 26. For example, the outer fastening member 62 may include a press-fit threaded insert 64 mounted to the back plate member 26 and a fastening screw 66 extending from the raised fastening support member 38 and into the press-fit threaded insert 64.

(19) The tensioning assembly 28 may also include a pair of inner fastening members 68. Each inner fastening member 68 may be positioned within a corresponding recessed channel 42 and extend outwardly from a corresponding dividing ridge 44 to the back plate member 26. For example, the inner fastening member 68 may include a press-fit threaded insert 64 coupled to corresponding dividing ridge 44 and a fastening screw 66 extending from the corresponding dividing ridge 44 and into the press-fit threaded insert 64.

(20) Referring to FIGS. 34-42, in the illustrated embodiment, the attachment system 14 is coupled to a rear surface of the back plate member 26 and is configured to facilitate mounting the ammunition magazine pouch 10 to a Molle mounting system and/or tactical belt. The attachment system 14 includes an attachment member 70 including an attachment body 72 having a substantially rectangular cross-sectional shape and a mounting strap 74 extending outwardly from a top end of the attachment body 72. In some embodiments, the attachment system 14 may include a

pair of mounting straps **74** extending outwardly from a top end of the attachment body **72**.

(21) The attachment system **14** may also include a snap button assembly **76** (shown in FIG. **14**) including a stud assembly **78** coupled near a bottom end of the attachment body **72** and a socket assembly **80** coupled to a distal end of the mounting strap **74**. The mounting strap **74** is adapted to fold along the top end of the attachment body **72** to fasten the corresponding socket assembly **80** to the stud assembly **78**. In some embodiments, the attachment member **70** is formed of a unitary body including a polymer-coated fabric composite material. The back plate member **26** may also include a stud opening **82** for coupling the stud assembly **78** to the back plate member **26** through the attachment body **72**.

(22) The attachment system **14** also includes a pair of locking tabs **84** coupled to the attachment body **72** and oriented substantially perpendicular to the mounting strap **74**. Each locking tab **84** extends between opposing ends that are coupled to the attachment body **72** to allow the mounting strap **74** to be positioned between each locking tab **84** and the attachment body **72** to facilitate mounting the ammunition magazine pouch **10** to a Molle mounting system and/or tactical belt.

(23) In some embodiments, the back plate member **26** may include a plurality of recessed stitching channels **86** (shown in FIG. **34**) defined along a front surface of the back plate member **26**. The recessed stitching channels **86** are sized and shaped to receive stitching for coupling the attachment body **72** to the support plate **50**.

(24) In some embodiments, the attachment system **14** includes the Hyperlock™ Attachment System provided by Safe Life Defense™. The Hyperlock™ Attachment System is an ultra-low profile, lightweight & durable assembly constructed of a Polymer-Coated Cordura® Fabric Composite that allows for easy mounting to nearly anything. Hyperlock™ securely attaches to traditional, micro and laser cut molle as well as slide on belts between 1.5"-2.25." The Hyperlock™ Attachment System includes dual locking tabs at the bottom of the pouch that are specific to the gear the system is mounting to. A top locking tab is used for Micro Molle and 1.5"-2" slide on belts, and a bottom locking tab is used for Traditional Molle and 2.25" slide on belts.

(25) In some embodiments, the front cover member **24** of the magazine support assembly **12** includes a plurality of sidewalls extending outwardly from the support flange **36** to form the magazine chamber **16** that is sized and shaped to receive an ammunition magazine therein. The front endwall **32** is spaced from the support flange **36** along a first axis (e.g., a Z-axis) and extends between the bottom endwall **34** and the left and right sidewalls **30**. The bottom endwall **34** is spaced a distance from the open top end **18** along a second axis (e.g., a Y-axis) that is perpendicular to the first Z-axis. The left sidewall is spaced a distance from the right sidewall along a third axis (an X-axis) that is perpendicular to the first Z-axis and the second Y-axis.

(26) In one embodiment, the open top end **18** is configured to receive the plurality of ammunition magazines. The bottom endwall **34** is configured to support and hold the at least one ammunition magazine within the ammunition magazine pouch **10**. The front cover member **24** includes a plurality of recessed channels **42** and a pair of raised fastening support members **38**. Each recessed channel **42** defines a corresponding magazine compartment **46** configured to receive and house at least one ammunition magazine. Each raised fastening support member **38** is configured to receive the tensioning assembly **28** to couple the front cover member **24** to the back plate member **26**.

(27) In various embodiments, the front cover member **24** includes a plurality of fastener openings **88** configured to receive the tensioning assembly **28**. The tensioning assembly **28** includes a plurality of adjustable fastening members **60**. Each of the plurality of fastener openings **88** on the front cover member **24** includes a cylindrical inner surface extending through the front endwall **32** and is sized and shaped to receive the adjustable fastening members **60** therein. In one embodiment, the front cover member **24** is coupled to the back plate member **26** via the one or more adjustable fastening members **60**. The adjustable fastening members **60** includes a press-fit threaded insert **64** that is coupled to the back plate member **26** and a fastening screw **66** inserted through a corresponding fastener openings **88** defined through the front cover member **24** and

threadably coupled to the press-fit threaded insert **64** to facilitate coupling the front cover member **24** to the back plate member **26**. In some embodiments, the fastening screw **66** includes a head diameter equal to 0.2090 inches. In other embodiments, the fastening screw **66** may include any suitable head diameter.

(28) The back plate member **26** includes a support plate **50**. The flexible attachment member **70** is coupled to a rear surface of the support plate **50** and includes a plurality of locking tabs **84** coupled to a rear surface of the flexible attachment member **70**. The back support plate **50** includes a substantially rectangular-shaped that includes the front surface **52** and the rear surface **54** along the first Z-axis. The support plate **50** extends between a bottom end, a top end, a left side, and a right side. The bottom end is spaced a distance from the top end along the second Y-axis. The left side is spaced a distance from the right side along the third X-axis. The back plate member **26** further includes at least one raised positioning members **56** having a substantially rectangular shape and extending outwardly from the front surface **52**. The raised positioning members **56** extends into the magazine chamber **16** defined between the front cover member **24** and the back plate member **26** and is sized and shaped to assist in holding the ammunition magazine in place in the ammunition magazine pouch **10**.

(29) The flexible attachment member **70** includes the attachment body **72** that includes a front surface **90**, a rear surface **92**, a top end, a bottom end, a left side, a right side, a pair of mounting straps **74** and a plurality of openings **94**. The front surface **90** is spaced a distance from the rear surface **92** along the first Z-axis. The top end is spaced a distance from the bottom end along the second Y-axis and the left side is spaced a distance from the right side along the third X-axis. The pair of mounting straps **74** extend out from the top end along the second Y-axis. The plurality of openings **94** are configured to receive the snap button assembly **76**. Each of the plurality of openings **94** on the flexible attachment member **70** includes a cylindrical inner surface extending through the front surface **90** and is sized and shaped to receive the snap button assembly **76** therein. The front surface **90** of the flexible attachment member **70** is coupled to the rear surface **54** of the back plate member **26**.

(30) In one embodiment, the pair of mounting straps **74** are configured to attach the ammunition magazine pouch **10** to a utility belt. Each mounting strap **74** includes at least one hole where each hole includes a cylindrical inner surface extending between the front surface **90** and the rear surface **92** and is sized and shaped to receive the socket assembly **80** therein. The pair of mounting straps **74** can then be bent to snap and secure the ammunition magazine pouch **10** to the utility belt.

(31) The pair of locking tabs **84** includes a top locking tab **96** and a bottom locking tab **98**. The top locking tab **96** includes a front side, a back side, a top end, a bottom end, a left side, and a right side. The front side is spaced a distance from the back side along the first Z-axis. The top end is spaced a distance from the bottom end along the second Y-axis and the left side is spaced a distance from the right side along the third X-axis. The bottom locking tab **98** includes a front side, a back side, a top end, a bottom end, a left side, and a right side. The front side is spaced a distance from the back side along the first Z-axis. The top end is spaced a distance from the bottom end along the second Y-axis and the left side is spaced a distance from the right side along the third X-axis. The back side of the top locking tab **96** and the back side of the bottom locking tab **98** are partially coupled to the rear surface **92** of the flexible attachment member **70**. The top locking tab **96** is spaced a distance from the bottom locking tab **98** along the second Y-axis. The top locking tab **96** and the bottom locking tab **98** are configured to assist in coupling the ammunition magazine pouch **10** to, for example, a utility belt. In one embodiment, shown in FIG. **14**, the bottom locking tab **98** includes a slit extending between the left and right side along the second Y-axis.

(32) Referring to FIGS. **43-62**, in some embodiments, the present invention describes a system **100** for manufacturing the tactical pouch **10**. The system **100** includes a thermoplastic machine assembly **102** and a Computerized Numerical Control (CNC) milling machine **104**. The thermoplastic machine assembly **102** includes a support frame **106**, a heat press machine **108**

mounted to the support frame **106**, a vacuum forming machine assembly **110** mounted to the support frame **106** and positioned adjacent to the heat press machine **108**, and a support sled **112** slideably coupled to the support frame **106**. The heat press machine **108** is configured to apply a heat and pressure treatment to a sheet of thermoplastic material to facilitate thermoforming the thermoplastic material.

(33) The vacuum forming machine assembly **110** includes a mold support frame **114** for supporting a mold assembly **116**, a hydraulic lifting assembly **118** coupled to the mold support frame **114**, and a vacuum pressure system **100**. The hydraulic lifting assembly **118** is configured to move the mold support frame **114** between a first vertical position (shown in FIG. **43**) and a second vertical position (shown in FIG. **44**). The vacuum pressure system **100** is coupled to the mold support frame **114** and configured to apply a vacuum pressure to the mold assembly **116** with the mold support frame **114** in the second vertical position.

(34) The support sled **112** is movable between the heat press machine **108** and the vacuum forming machine assembly **110**, and is configured to receive a sheet of thermoplastic material **120** therein to move the sheet of thermoplastic material **120** between the heat press machine **108** and the vacuum forming machine assembly **110**. The support sled **112** is positionable between a first sled position (shown in FIG. **46**) with the sheet of thermoplastic material positioned within the heat press machine **108**, and a second sled position (shown in FIG. **45**) with the sheet of thermoplastic material positioned above the vacuum forming machine assembly **110**.

(35) In the illustrated embodiment, the support sled **112** includes a slide rail assembly **122** mounted to the support frame **106**, a base sled frame **124** mounted to the slide rail assembly **122**, and a cover frame **126** that is pivotably mounted to the base sled frame **124**. The cover frame **126** is positionable between an open position (shown in FIG. **44**) and a closed position (shown in FIG. **45**).

(36) The base sled frame **124** includes a frame member **128**, a support flange **130**, and a first sealing gasket **132** mounted on the support flange **130**. The frame member **128** defines a mold opening having a substantially rectangular cross-sectional shape. The mold opening is sized and shaped to receive the mold assembly **116** therein. The support flange **130** extends inwardly from the frame member **128** about a perimeter of the mold opening.

(37) The support sled **112** is configured to support a sheet of thermoplastic material **120** between the first sealing gasket **132** of the base sled frame **124** and the cover frame **126** with the cover frame **126** in the closed position.

(38) In the illustrated embodiment, the mold support frame **114** includes a second sealing gasket **134** positioned about a perimeter of the mold assembly **116**. The second sealing gasket **134** of the mold support frame **114** contacts the first sealing gasket **132** of the base sled frame **124** when the mold support frame **114** is moved to the second vertical position to facilitate a vacuum seal being created between the mold assembly **116** and the sheet of thermoplastic material **120**.

(39) During operation, the vacuum pressure system **100** applies the vacuum pressure to the mold assembly **116** with the sheet of thermoplastic material **120** supported between the first sealing gasket **132** of the base sled frame **124** and the cover frame **126**, and with the second sealing gasket **134** of the mold support frame **114** in contact with the first sealing gasket **132** of the base sled frame **124**.

(40) In some embodiments, the mold assembly **116** includes an array **136** of positive front cover mold assemblies **138**. The positive front cover mold assembly **138** includes a base member **140** and a cover member **142** extending outwardly from the base member **140**. The cover member **142** includes a plurality of sidewalls **144**, **146**, **148** extending between a first endwall **150** and an opposite second endwall **152**. The plurality of sidewalls includes a top sidewall **146** and a pair of opposing sidewalls **144**, **148** extending outwardly from the top sidewall **146** towards the base member **140**. A pair of opposing raised members **154** are positioned adjacent each opposing sidewall **144**, **148**.

(41) The mold assembly **116** may also include an array **156** of positive back plate mold assemblies

158. The positive back plate mold assembly **158** includes a base member **160** and a pair of opposing back plate mold members **162** that extend outwardly from the base member **160**. The pair of opposing back plate mold members **162** are spaced a distance apart and arranged in a mirrored relationship. In the illustrated embodiment, each back plate mold member **162** includes a planar outer surface **164** and a pair of raised positioning members **166** extending outwardly from the planar outer surface **164**. Each raised positioning member **166** includes a top endwall extending between a pair of sidewalls, a first endwall, and a second endwall opposite the first endwall. The second endwall extends from the top endwall towards the planar outer surface **164** at an oblique angle.

(42) The CNC milling machine **104** includes a table **168**, a CNC milling assembly **170** mounted above the table **168**, a vacuum mold assembly **172** mounted on the table, and a vacuum pressure system **100** coupled to the vacuum mold assembly **172**. The vacuum mold assembly **172** includes a support plate **174** and a support mold **174** mounted to the support plate **174**. The support mold **174** includes a base member **176**, a plurality of positioning members **178** extending outwardly from the base member **176**, a plurality of openings **180** extending through the base member **176**, and a plurality of silicone gaskets **182** mounted on the base member **176**. The vacuum pressure system **100** is configured to apply a vacuum pressure to the vacuum mold assembly **172** through the plurality of openings **180** to facilitate mounting a hardened thermoplastic assembly to the support mold **174** during a milling operation.

(43) In some embodiments, the system **100** may include a stitching mold assembly **184** that includes stitching base mold **186** and a stitching guide mold **188**. The stitching base mold **186** includes a recessed area **190** sized and shaped to receive the attachment member **70** therein, and one or more recessed channels **192** defined within the recessed area **190** that are sized and shaped to receive a corresponding locking tab **84** therein. The stitching guide mold **188** includes a stitching guide mold body including a plurality of stitching guide slots **194** that are defined along the stitching guide mold body.

(44) In some embodiments, the present invention includes a molding assembly **116** for forming the ammunition magazine pouch **10**. The molding assembly **116** includes a plurality of molds to help form the different components of the ammunition magazine pouch **10**. The plurality of molds includes a back form cutout mold **158**, a back form full array mold **156**, a front form mold **138**, a brother pallet clamp mold **184**, a pallet clamp base mold **186**, and a pallet clamp top mold **188**. The plurality of molds are configured to help form the different components of the ammunition magazine pouch **10**. For instance, the back form cutout mold **158** is configured to form the back support member **26** by pressing the material over the mold and cutting each individual component of the ammunition magazine pouch **10**. The material used to make the ammunition magazine pouch **10** is passed under a specific tool and pressed against the plurality of molds to make each part.

(45) In one embodiment, the back form cutout mold **158** is configured to make the back support member **26**. The back form full array mold **156** is configured to form a plurality of back support members. Similar to the back form cutout mold **158**, the front form mold **138** is configured to make a plurality of magazine front cover members **24** where the material used to make each magazine front cover member **24** is pressed or molded against the front form mold.

(46) In another embodiment, the brother pallet claim mold, the pallet clamp base mold, and the pallet clamp top mold each include a plurality of holes. The plurality of holes are used as a template to drill the plurality of fastener openings for receiving the fastening assembly on both the magazine support member and the back plate assembly. For example, the pallet clamp base mold is placed on the finished flexible fastening member to layout where the plurality of openings need to be cut. Once each of the components of the ammunition magazine pouch **10** are formed each opening is sized and cut to receive a corresponding bolt set or snap set from the fastening assembly.

(47) FIG. **63** is a flowchart illustrating a method **200** of manufacturing the tactical pouch **10**. The method includes a plurality of steps. Each method step may be performed independently of, or in

combination with, other method steps. Portions of the methods may be performed by any one of, or any combination of, the components of the system **100**.

(48) Method step **202** includes thermoforming a front cover member **24** from a first sheet of thermoplastic material **120**. For example, the first sheet of thermoplastic material **120** is placed within the support sled **112** with the support sled **112** in the second sled position and with the mold support frame **114** positioned in the first vertical positions such that the mold assembly **116** does not contact the sheet of thermoplastic material **120**. The first sheet of thermoplastic material **120** is placed onto the base sled frame **124** and supported by the support flange **130** and a first sealing gasket **132** about a perimeter of the about a perimeter of the first sheet of thermoplastic material **120**. The cover frame **126** is then moved to the closed position and secured with a locking clamp. The support sled **112** is then moved to the first sled position to position the first sheet of thermoplastic material **120** within the heat press machine **108**.

(49) The heat press machine **108** is then operated to apply a heat treatment to the first sheet of thermoplastic material **120** to form a malleable thermoplastic sheet. For example, the heat press machine **108** is operated to apply the heat treatment at a predefined temperature and predefined pressure for a predefined period of time. At the completion of the heat treatment, the support sled **112** is then moved from the first sled position to the second sled position to position the malleable thermoplastic sheet over the mold assembly **116** including the array **136** of positive front cover mold assemblies **138**. The hydraulic lifting assembly **118** then operates to move the mold support frame **114** and mold assembly **116** from the first vertical position to the second vertical position to place the second sealing gasket **134** of the mold support frame **114** in contact with the first sealing gasket **132** of the base sled frame **124**. The vacuum pressure system **100** then applies a vacuum pressure to the malleable thermoplastic sheet to form a molded front cover assembly over the positive front cover mold assembly **138**. A fan is then operated to facilitate cooling the molded front cover assembly to form a hardened front cover assembly.

(50) Once the hardened front cover assembly is formed, the vacuum pressure system **100** is operated to release the vacuum pressure, the hydraulic lifting assembly **118** is operated to move the mold support frame **114** and mold assembly **116** from the second vertical position to the first vertical position, and the hardened front cover assembly is removed from the support sled **112** and placed over the vacuum mold assembly **172** of the CNC milling machine **104**. The vacuum pressure system **100** of the CNC milling machine **104** is then operated to apply a vacuum pressure to the vacuum mold assembly **172** to facilitate mounting the hardened front cover assembly to the support mold **174**. The CNC milling machine **104** is then operated to mill excess material from the hardened front cover assembly and drill a plurality of fastener openings through the hardened front cover assembly for receiving the adjustable fastening members **60** of the tensioning assembly **28**. The milled hardened front cover assembly is then cut in half to form two front cover members **24**.

(51) Method step **204** includes thermoforming a back plate member **26** from a second sheet of thermoplastic material **120**. Similar to the process of method step **202**, the second sheet of thermoplastic material **120** is placed within the support sled **112** with the support sled **112** in the second sled position and with the mold support frame **114** positioned in the first vertical position. The cover frame **126** is then moved to the closed position and the support sled **112** is moved to the first sled position to position the second sheet of thermoplastic material **120** within the heat press machine **108**.

(52) The heat press machine **108** is then operated to apply a heat treatment to the second sheet of thermoplastic material **120** to form a second malleable thermoplastic sheet. At the completion of the heat treatment, the support sled **112** is then moved from the first sled position to the second sled position to position the second malleable thermoplastic sheet over the mold assembly **116** including the array **156** of positive back plate mold assemblies **158**. The hydraulic lifting assembly **118** then operates to move the mold support frame **114** and mold assembly **116** from the first vertical position to the second vertical position to position the malleable thermoplastic sheet onto a positive

back plate mold assembly **158**. The vacuum pressure system **100** then applies a vacuum pressure to the malleable thermoplastic sheet to form a molded back plate assembly over the positive back plate mold assembly **158**, and a fan is operated to facilitate cooling the molded back plate assembly to form a hardened back plate assembly.

(53) The hardened back plate assembly is then removed from the support sled **112** and placed over the vacuum mold assembly **172** of the CNC milling machine **104**. The vacuum pressure system **100** of the CNC milling machine **104** is then operated to apply a vacuum pressure to the vacuum mold assembly **172** to facilitate mounting the hardened back plate assembly to the support mold **174**. The CNC milling machine **104** is then operated to mill excess material from the hardened back plate assembly and drill a plurality of fastener openings and stud openings through the hardened front cover assembly for receiving the adjustable fastening members **60** of the tensioning assembly **28** and the attachment system **14**. The CNC milling machine **104** is also operated to etch the recessed stitching channels **86** along the front surface of the back plate member **26** for receiving the stitching for coupling the attachment system to the back plate member **26**. The milled hardened back plate assembly is then cut in half to form two back plate members **26**.

(54) Method step **206** includes stitching the back plate member **26** to an attachment system **14**. For example, the attachment system **14** is positioned onto the stitching base mold **186** with the locking tabs **84** positioned in the recessed channels **192** and the attachment member **70** positioned over the locking tabs **84** within the recessed area **190**. The back plate member **26** is then positioned onto the attachment system **14** within the recessed area **190**. The stitching guide mold **188** is then positioned onto the back plate member **26**, and a stitching machine is operated to apply stitching to the attachment system **14** and the back plate member **26** along the plurality of stitching guide slots **194**. The recessed stitching channels **86** of the back plate member **26** correspond to the stitching guide slots **194** to facilitate reducing stitching protruding into the magazine chamber **16**. The snap button assembly **76** is then coupled to the attachment member **70** and the back plate member **26**.

(55) Method step **208** includes coupling the front cover member **24** to the back plate member **26** to define the magazine chamber **16** therebetween and extending between the open top end **18** and the closed bottom end **20**. For example, the front cover member **24** may be coupled to the back plate member **26** with an adhesive such that the support flange **36** is adhered to the front surface of the back plate member **26**.

(56) Method step **210** includes coupling the tensioning assembly **28** to the front cover member **24** and the back plate member **26** to form the tactical pouch **10**. In some embodiments, press-fit threaded inserts are mounted to the back plate member **26** before adhesive is applied and the front cover member **24** is coupled to the back plate member **26**. The fastening screws are then inserted through the fastener openings and into the press-fit threaded inserts.

(57) Obviously, many modifications and variations of the present invention are possible in light of the above teachings. The invention may be practiced otherwise than as specifically described within the scope of the appended claims.

(58) Although specific features of various embodiments of the disclosure may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the disclosure, any feature of a drawing or other embodiment may be referenced and/or claimed in combination with any feature of any other drawing or embodiment.

(59) This written description uses examples to describe embodiments of the disclosure and also to enable any person skilled in the art to practice the embodiments, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. An ammunition magazine pouch comprising: a magazine support assembly including an interior surface defining a magazine chamber extending between an open top end and a closed bottom end, the magazine chamber configured to receive at least one ammunition magazine therein, the magazine support assembly including: a front cover member formed of a unitary body including: an inner surface defining a portion of the magazine chamber; a pair of opposing sidewalls extending outwardly from a front endwall; a bottom endwall extending between the front endwall and the opposing sidewalls; a support flange extending outwardly from the opposing sidewalls and the bottom endwall defining a perimeter of the front cover member; and a pair of recessed channels defined along an outer surface of the front cover member and extending from the front endwall to the bottom endwall, each recessed channel defines a corresponding dividing ridge extending into the magazine chamber to define a plurality of magazine compartments within the magazine chamber, each magazine compartment configured to receive a corresponding ammunition magazine therein; a back plate member coupled to the front cover member to define the magazine chamber therebetween, the back plate member including a support plate having a substantially rectangular cross-sectional shape extending between a front surface and a rear surface, the front surface coupled to the support flange of the front cover member with an adhesive and including a pair of raised positioning members extending into the magazine chamber, each raised positioning member orientated with respect to a corresponding dividing ridge of the front cover member to define the plurality of magazine compartments; and a tensioning assembly including a plurality of adjustable fastening members spaced along an upper portion of the front cover member and coupled between the front cover member and the back plate member, and configured to adjust a distance between the upper portion of the front cover member and an upper portion of the back plate member to adjust a width of the open top end of the magazine chamber; and an attachment system coupled to the rear surface of the back plate member and including: an attachment member including an attachment body and a pair of mounting straps extending outwardly from a top end of the attachment body, the attachment body having a cross-sectional shape similar to the substantially rectangular cross-sectional shape of the support plate of the back plate member; and a pair of locking tabs coupled to the attachment body and oriented substantially perpendicular to the pair of mounting straps; wherein the back plate member includes a plurality of recessed stitching channels defined along the front surface of the back plate member and configured to receive stitching for coupling the attachment body to the support plate.
2. The ammunition magazine pouch of claim 1, wherein the front cover member includes: a pair of opposing raised fastening support members extending outwardly from the opposing sidewalls defined near the open top end of the magazine support assembly.
3. The ammunition magazine pouch of claim 2, wherein the plurality of adjustable fastening members includes a pair of opposing outer fastening members coupled to the opposing raised fastening support members and the back plate member.
4. The ammunition magazine pouch of claim 3, wherein each raised fastening support member includes an inner surface spaced a distance from the back plate member to define a gap between the inner surface of the raised fastening support member and the back plate member.
5. The ammunition magazine pouch of claim 4, wherein each outer fastening member includes a press-fit threaded insert mounted to the back plate member and a fastening screw extending from the raised fastening support member and into the press-fit threaded insert.
6. The ammunition magazine pouch of claim 2, wherein the front cover member includes a plurality of drain openings spaced along the bottom endwall.
7. The ammunition magazine pouch of claim 2, wherein the front cover member is formed of a thermoplastic material.

8. The ammunition magazine of claim 1, wherein the attachment system includes: a snap button assembly associated with each mounting strap, each snap button assembly including a stud assembly coupled near a bottom end of the attachment body and a socket assembly coupled to a distal end of a corresponding mounting strap, each mounting strap adapted to fold along the top end of the attachment body to fasten the corresponding socket assembly to a corresponding stud assembly.
 9. The ammunition magazine of claim 8, wherein the back plate member includes a pair of stud openings for coupling each stud assembly to the back plate member and through the attachment body.
 10. The ammunition magazine pouch of claim 9, wherein each locking tab extends between opposing ends that are coupled to the attachment body to allow each mounting strap to be positioned between each locking tab and the attachment body to facilitate mounting the ammunition magazine pouch to a Molle mounting system and/or tactical belt.
 11. The ammunition magazine pouch of claim 1, wherein the attachment member is formed of a unitary body including a polymer-coated fabric composite material.
 12. The ammunition magazine pouch of claim 1, wherein the tensioning assembly includes a pair of inner fastening members, each inner fastening member positioned within a corresponding recessed channel and extending outwardly from a corresponding dividing ridge to the back plate member.
 13. The ammunition magazine pouch of claim 12, wherein each raised positioning member is spaced a distance from the corresponding dividing ridge, each inner fastening member includes a press-fit threaded insert coupled to corresponding dividing ridge and a fastening screw extending from the corresponding dividing ridge and into the press-fit threaded insert.
 14. The ammunition magazine pouch of claim 13, wherein the support plate includes an arcuate surface extending along a top portion of the support plate to define the open top end of the magazine chamber.
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